

Selected Component	Alternative	Comparison	Reason for Preference
ESP-WROOM-32 (Microcontroller)	Arduino Uno / Raspberry Pi Pico	ESP32 is faster (240MHz vs 16MHz), built-in Wi-Fi & Bluetooth, lower power use.	Best for smart robots: wireless + vision + cost-effective.
GM25 370 Motor (Driving Motor)	N20 Gear Motor / DC 775	Medium size, good torque, lower consumption than DC 775.	Balanced torque, size, and price; fits LEGO wheels.
EV3 Medium Motor (Steering Servo)	MG995 / MG996R Servo	EV3 motor offers precise angle + speed control, LEGO-compatible, longer life.	High steering accuracy, easy LEGO integration.
L298N Motor Driver	L293D / BTS7960	L298N handles 2A, easy libraries, cheap and widely available.	Enough for GM25 motors, simple wiring, no complexity.
PiCy2 Camera (Object Tracking)	OpenMV Cam / Arducam	Built-in color & shape tracking in hardware, no external processing.	Reduces ESP32 load, faster response for robotics projects.
Ultrasonic HC-SR04	VL53L0X (ToF Sensor)	Longer range, easier use, more available than VL53L0X.	Simple, cheap, and best for obstacle detection.
MPU6050 (Gyro + Accelerometer)	BNO055 / MPU9250	MPU6050 is cheaper, widely available, good enough for balancing.	Low cost, sufficient for navigation.
4 × Spike Prime Wheels Ø56	Generic Rubber Wheels / RC Wheels	LEGO Technic compatible, lightweight, good surface grip.	Excellent traction + easy mounting.
LEGO Technic Pieces	3D Printed Parts / Metal Frames	Flexible, precise, easy to assemble/disassemble, safe to use.	Faster and simpler prototyping compared to custom-made parts.